

## **Client Portal Development Challenge Statement**

For Ngee Ann Polytechnic – School of InfoComm Technology

### **Industry Project / Applied Learning Collaboration**

**Industry Context: Fund Management / Financial Services**

**Project Title: Secure Investor Client Portal for Fund Management**

#### **1. Overview**

In the fund management industry, transparency, trust, and timely access to accurate performance information are critical to maintaining investor confidence. As investment operations scale, reliance on manual reporting and ad-hoc communication becomes increasingly unsustainable.

This project challenges student teams to design and develop a secure, investor-facing Client Portal that serves as a single digital touchpoint for authenticated investors to access account information and portfolio performance. The solution must project credibility, ensure data security, and support core investor self-service functions.

#### **2. Project Objectives**

The primary objective of this project is to deliver a minimum viable yet production-ready client portal that supports essential investor interactions while adhering to security and usability expectations.

Specifically, the solution should:

Provide secure authentication and controlled access to investor accounts

Present clear, accurate, and timely portfolio performance information

Lay a scalable foundation for future operational features

### 3. Scope of Work

Students are required to design and implement the following features, aligned to the Product Requirements Document (PRD).

#### **Compulsory Deliverables**

##### 3.1 Authentication and Account Management (Mandatory)

The portal must implement a robust authentication and account management framework that ensures only authorised users can access sensitive financial information.

Key requirements include:

Invitation-based investor onboarding (no public self-registration)

Secure login using email and password credentials

Support for Multi-Factor Authentication (MFA)

Role-based access control (e.g. Investor, Operations, Admin)

Session management, password reset, and account security safeguards

Audit logging of authentication-related activities

The design should reflect financial-industry best practices in security, access control, and user accountability.

##### 3.2 Dashboard and P&L Reporting (Mandatory)

The investor dashboard is the core user interface of the portal and must present portfolio performance information clearly and intuitively.

Key requirements include:

A personalised investor dashboard displaying account value and performance metrics

Daily Profit & Loss (P&L) reporting with clear date references

Visualisation of performance trends (e.g. charts, summaries)

Structured presentation of returns (e.g. daily, monthly, year-to-date)

A reliable data ingestion pipeline to support daily P&L updates

The dashboard should prioritise clarity, accuracy, and usability, enabling investors to quickly understand portfolio performance without requiring manual interpretation.

## **Optional Deliverables**

### **3.3 Deposit and Withdrawal Requests (Optional / Bonus Scope)**

As an extension to the core portal, teams may implement functionality that allows investors to submit deposit and withdrawal requests digitally.

This optional feature may include:

Submission of deposit or withdrawal requests via the portal

Status tracking of requests (e.g. submitted, approved, completed)

Role-based review and approval workflows for operations staff

Clear separation between request tracking and actual fund movement

While optional, successful implementation of this feature will demonstrate deeper understanding of operational workflows and system design in regulated environments.

### **3.4 Support Multiple Fund Products and Specification of Risk Tolerance**

For the MVP, teams may assume that the investment firm only offers one product. However, in reality, the firm's product offerings can be more complex.

Teams may think about how to redesign overall data architecture and client portal to support the offering of multiple funds (i.e. client can choose to deposit funds into one or more fund offerings), and / or the ability for clients to specify their risk tolerance for each fund (i.e. for each fund, the investor can specify a risk tolerance level, which will limit how aggressively the fund invests).

### **3.5 Support Performance Analysis**

The fund portfolio manager is continuously reviewing fund performance to identify areas for optimisation. The performance review process can be made more effective if

the portfolio manager is able to break down his overall return into different drivers of return, across different time periods.

This would enable the portfolio manager to develop a better understanding of why the portfolio performed in a certain manner and perform what-if analysis to test out how the fund would have performed if investor capital had been allocated differently.

#### 4. Suggested Technology Platforms

To ensure consistency, feasibility, and alignment with industry-relevant practices, students are encouraged to build their solutions using one or more of the following suggested platforms. Teams may propose alternatives, subject to lecturer approval.

Frontend (Web Application):

React.js or Next.js

Vue.js (optional alternative)

Backend / Application Layer:

Node.js (Express / NestJS)

Python (FastAPI / Flask)

Authentication & Security:

JWT-based authentication

OAuth-compatible identity services (e.g. Azure AD B2C, Auth0 – simulated if needed)

Database:

PostgreSQL or MySQL (relational)

MongoDB (for selected components, if justified)

Cloud / Hosting (Simulated or Academic Tier):

Microsoft Azure (preferred, aligned with NP standards)

AWS or equivalent academic cloud environments

The focus of evaluation will remain on system design, security awareness, and functional correctness, rather than mastery of a specific framework.

## 5. Data Structure

A functioning client portal has to be supported by robust data architecture that supports critical day-to-day operations such as investor management, fund management, and PNL reporting.

The database should contain the following datasets:

### 5.1 Investor Accounts Dataset

| # | Field           | Description                                                     |
|---|-----------------|-----------------------------------------------------------------|
| 1 | Investor ID     | Unique identifier for each investor                             |
| 2 | Investor name   | Investor name                                                   |
| 3 | Onboarding date | Date first onboarded                                            |
| 4 | Role            | Investor; Operations (read-write operations); Admin (read-only) |

### 5.2 Fund Flows Dataset

| # | Field            | Description                                             |
|---|------------------|---------------------------------------------------------|
| 1 | Investor ID      | Unique identifier for each investor                     |
| 2 | Request ID       | Unique identifier for each deposit / withdrawal request |
| 3 | Request type     | Deposit / withdrawal                                    |
| 4 | Request date     | Date of request submission                              |
| 5 | Processed date   | Date request is processed                               |
| 6 | Request amount   | Amount to deposit or withdraw                           |
| 7 | Request currency | Currency denomination of request amount                 |

### 5.3 Portfolio Holdings Dataset

| # | Field               | Description                         |
|---|---------------------|-------------------------------------|
| 1 | Investor ID         | Unique identifier for each investor |
| 2 | As of date          | As of date                          |
| 3 | Fund 1 ID           | Fund unique identifier              |
| 4 | Fund 1 shareholding | % share of Fund 1's net asset value |
| 5 | Fund 2 ID           | Fund unique identifier              |

|   |                     |                                     |
|---|---------------------|-------------------------------------|
| 6 | Fund 2 shareholding | % share of Fund 2's net asset value |
|---|---------------------|-------------------------------------|

#### 5.4 Investment Transactions Dataset

| #  | Field       | Description                                                                                    |
|----|-------------|------------------------------------------------------------------------------------------------|
| 1  | ticket      | Unique deal ticket number assigned by the trading server                                       |
| 2  | order       | The ticket number of the order that triggered this deal                                        |
| 3  | time        | Deal execution time as a Unix timestamp (seconds)                                              |
| 4  | time_msc    | Deal execution time in milliseconds since Unix epoch                                           |
| 5  | type        | Deal type Buy, 1 = Sell, 2 = Balance                                                           |
| 6  | entry       | Deal entry direction, In (open), 1 = Out (close)                                               |
| 7  | magic       | Strategy ID                                                                                    |
| 8  | reason      | Reason or origin of the deal                                                                   |
| 9  | position_id | ID of the position this deal belongs to or created; links deals to a common position lifecycle |
| 10 | volume      | Traded volume (lot size) of the deal                                                           |
| 11 | price       | Execution price at which the deal was carried out                                              |
| 12 | commission  | Commission charged for the deal (negative value, in account currency)                          |
| 13 | swap        | Swap accrued on the deal at position close (in account currency)                               |
| 14 | profit      | Financial result of the deal in the account currency (realised P&L for closing deals)          |
| 15 | fee         | Any additional fees charged for the deal (in account currency)                                 |
| 16 | sl          | Stop Loss price level set at the time the deal was executed                                    |
| 17 | tp          | Take Profit price level set at the time the deal was executed                                  |
| 18 | symbol      | The trading instrument/symbol the deal was executed on (e.g. EURUSD)                           |
| 19 | comment     | Comment attached to the deal                                                                   |
| 20 | external_id | Deal ID in an external trading system (e.g. an exchange), if applicable                        |

Datasets 5.1, 5.2, and 5.3 do not exist yet – teams should consider whether the suggested data structure is sufficient for daily operations and the types of data that will need to be stored in each dataset.

Teams will be provided with dummy data for dataset 5.4. The data structure in dataset 5.4 does not match the data structure provided as it is raw output provided by the firm's

broker. Teams should consider how best to store the data and decide which fields are necessary to meet project requirements.



5.4 Dummy Data.csv

All datasets will be fictional and contain no real client or financial data, but will be structured to reflect industry-realistic schemas and constraints.

## 6. Expected Outcomes

By the end of the project, student teams should be able to:

Translate a formal PRD into a coherent system design

Implement secure, role-based web applications handling sensitive data

Demonstrate awareness of financial services expectations, including security, auditability, and professionalism

Deliver a functional prototype that can be extended beyond the academic context

## 7. Evaluation Focus

Student solutions will be assessed based on:

Alignment with compulsory feature requirements (Sections 4.1 and 4.2)

Security awareness and access control design

Clarity and usability of the investor dashboard

Code quality, documentation, and architectural soundness

Thoughtful handling of optional scope (if attempted)

This challenge statement is intended to mirror real-world industry problem framing and expectations, preparing students for professional software delivery in regulated sectors.

## 8. Appendix

This project requires students to understand and implement two different concepts:

### 8.1 Calculating investor fund shareholding

An open-ended fund allows various investors to make deposits and withdrawals at different points in time. It is therefore important for the system to keep track of each investor's net deposits (and their share of fund profit & loss).

A simplified recordkeeping process is as follows:

First assume the fund begins day 1 with a certain amount of total assets (denoted as  $x$ ), with each investor owning a share of  $x$ .

At the end of day 1, the fund would have (i) generated PNL and also (ii) received a number of deposit / withdrawal requests.

Day 1 ending balance = Total assets at beginning of Day 1 (i.e.  $x$ ) + PNL generated (i)

At this point in time, each investor's % shareholding has not changed, as the PNL is split among investors in the same proportion as their shareholding at the start of Day 1.

We then calculate the dollar value of each investor's share by multiplying % shareholding by Day 1 ending balance.

Next, we process all deposit and withdrawal requests. Investors that submitted withdrawal requests will see the dollar value of their share decrease, and vice versa. Finally, we then recalculate the % shareholding for each investor.

What we end up with is Day 2 starting balance. The process then repeats for Day 2, 3, and so on.

### 8.2 Measuring investment performance

When measuring performance, one should minimally provide the following two types of metrics: (i) total dollar PNL, and (ii) fund return

Also take note that performance measures are also time period dependent, so a start date and end date will have to be provided as input.

(i) Total dollar PNL for each investor can be easily calculated if the intermediate workings stored in table 8.1 are stored properly. Total dollar PNL would then be the sum of PNL generated \* investor's share of PNL on each day.



(ii) Fund return would simply be the daily % fund return compounded from start date till end date. Daily % fund return can be computed as  $\text{daily PNL} / \text{total assets at beginning of day}$